PLANNING, ORGANISING & RISK CONTROL

How our team of 7 builds the Car Rental Management System: who does what, in which sprint, with which tools, and what we do when things go wrong. Every task and date comes from our Trello board.

7 PEOPLE | 8 SPRINTS | 3 MILESTONES | 10 TRACKED RISKS

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Attached inside this PDF: **Wayfare-Schedule.gan** (GanttProject) and **Wayfare-Planning.xlsx** (Excel). Open your PDF reader's attachments panel to save them.

01 · OVERVIEW AND SCOPE

What we are building, and what we are not

The project is a web-based Car Rental Management System. Customers browse cars, book them and follow their rentals. Employees manage the fleet, approve requests, hand over keys and record returns. Every task, name and date in this plan comes from our Trello board, so the team shares one picture of the work, the order and the risks.

1
Working system
All eight features work end to end with the front end, from registration to payments.

2
On time
Every sprint ends on its date. The last feature lands on 29 Nov and testing ends on 25 Dec.

3
No double bookings
A car is never booked twice for overlapping dates.

4
Easy to use
A first-time customer can book a car without any help.

In scope

- **Customer side:** register, log in, list cars for chosen dates, request a rental, cancel it, my rentals, profile.
- **Staff side:** lists of customers, cars, rentals and payments; approve, reject, hand over keys, record returns; add, edit and delete cars.
- **Back end:** database, a controller with DTOs for every endpoint, login and roles, rules such as no overlapping bookings.
- **Quality:** testing with the front end and bug fixing in every sprint, then a final testing sprint.

Out of scope

- Online card payments. The system keeps payment records and applies the refund rules when staff reject a request.
- A native mobile app. The site works on phones in the browser.
- Multi-branch fleets, GPS tracking, insurance quotes.
- Anything outside the agreed endpoint list, unless a change request is approved.

STACK: ASP.NET CORE WEB API · ENTITY FRAMEWORK CORE · SQL SERVER · IDENTITY + JWT · HTML, CSS, JAVASCRIPT · GITHUB

DELIVERABLES

DELIVERABLE	WHAT IT CONTAINS	DUE
Project proposal	Two-page essay: intended use and overall functionality.	11 Oct
Project management plan	This document: team, sprints, Gantt chart, risks, tools.	11 Oct
Front-end wireframe	14 linked pages with sample data, reviewed by the team. Sprint 1.	Done
Data layer	Entities, ERD, models and enums, endpoint list, DbContext and database. Sprint 2.	13 Oct
Customer API	Login, register, car list, rental request, cancel, my rentals, profile. Sprints 3 and 4.	2 Nov
Staff API	Customer, car, rental and payment lists; approve, reject, hand over keys, record returns; add, edit and delete cars. Sprints 5 to 7.	2 Dec
Tested system	The whole system tested with the front end and the bugs fixed. Sprint 8.	25 Dec

Constraints we plan around: a team of 7 students working part time next to other courses, fixed course deadlines, and no budget. All tools in section 06 are free.

02 · TEAM

Seven people, one board

We are seven students. All of us are back-end developers, and we built the front end together as well. We have not set fixed roles: the work is split into Trello cards, and each card carries the names of the people working on it.

01	Abdelrahman Hany	TEAM LEADER
02	Kerollos Romany	TEAM MEMBER
03	Mina Maher	TEAM MEMBER
04	Samuel Mokhles	TEAM MEMBER
05	Matthew Ashraf	TEAM MEMBER
06	Marwan Mohamed	TEAM MEMBER
07	Basem Hany	TEAM MEMBER

How we use the board

- **One list per sprint.** Eight lists, Sprint 1 to Sprint 8. Every task is a card with a start date and a due date.
- **Names on cards.** Whoever works on a card adds their name. Shared work, such as requirements and bug fixing, carries the whole team.
- **Sprint hand-out.** The cards of the next sprint get their names when that sprint starts, in the Monday meeting.
- **Done means ticked.** A card is done when its due date is ticked complete. Labels mark features, testing, front-end integration, database, authentication and milestones.

SCHEDULE
The team leader decides, after hearing everyone.

TECHNICAL CHOICES
The people working on it decide. Disagreements go to the weekly meeting.

SCOPE CHANGES
Only through a change request, approved by the team.

TEAM AGREEMENT

Reply within a day
Messages in the group chat get an answer within 24 hours.

Show up or say so
If you cannot join a meeting, tell the team leader beforehand and read the notes.

Small, clear commits
Commit often with a short message that says what changed and why.

Flag blockers early
Stuck for more than a day? Say it in the mid-week check-in, not at the due date.

03 · WORK BREAKDOWN

Eight sprints, every card on the board

The board splits the work into eight sprints of about ten days, one Trello list each. Every row below is one card, in date order, with its start and due dates. Green rows are done. From Sprint 3 on, names are added to the cards when the sprint starts.

SPRINT 1 — DONE

Front-end wireframe and project structure

23 Sep – 3 Oct · 11 days · 6 cards

✓ 1.1	Set up GitHub and add project members	Kerollos	23–24 Sep
✓ 1.2	Project requirements	Whole team	24–27 Sep
✓ 1.3	Choose the web stack	Whole team	27 Sep
✓ 1.4	Determine the project structure	Whole team	27 Sep
✓ 1.5	Front-end wireframe	Abdelrahman	27–30 Sep
✓ 1.6	Front-end review and notes	Mina, Kerollos	30 Sep – 3 Oct

SPRINT 2 — DONE

Back-end data and database

3–13 Oct · 11 days · 9 cards

✓ 2.1	Business flow and requirements	Whole team	3 Oct
✓ 2.2	Entities and their attributes	Kerollos	3–4 Oct
✓ 2.3	Install the required NuGet packages	Mina, Kerollos	4 Oct
✓ 2.4	ERD and its documentation	Mina	5–7 Oct
✓ 2.5	Back-end models and enums	Mina	7–9 Oct
✓ 2.6	List all API endpoints	Mina	9–10 Oct
✓ 2.7	DbContext and database ◆ M1	Kerollos	9–11 Oct
✓ 2.8	Bug fixing and edits	Whole team	3–13 Oct
✓ 2.9	Back-end data configurations	Kerollos	11–13 Oct

How to read the board

- **Lists are sprints,** cards are tasks. A card runs from its start date to its due date.
- **Controllers and DTOs.** Each endpoint has a controller card, and half of them also have a card for the DTO and its mapper.
- **Every sprint tests.** Sprints 3 to 7 end with a “Testing with the front end” card and a “Bug fixing and edits” card.
- **Start dates.** Cards added on 15 Oct only have a due date on Trello. Their planned start is in the card text and is used here.
- **Same list in Excel.** The attached workbook has every card, with a link to it on Trello.

Labels on the cards

- Feature
- Database
- Authentication
- Integrating with Frontend
- Testing
- ◆ M1 Milestone
- ✓ Due date ticked
- Done row

03 · WORK BREAKDOWN, CONTINUED

Sprints 3 and 4: sign in and the customer side

SPRINT 3 — IN PROGRESS · 0 OF 17 DONE

Authentication endpoints and one customer endpoint

13–22 Oct · 10 days · 17 cards · names added at sprint start

○	3.1	Login controller	13–15 Oct
○	3.2	Register controller	13–15 Oct
○	3.3	Register DTO and its mapper	13–15 Oct
○	3.4	Login DTO and its mapper	13–15 Oct
○	3.5	Fix enum typos and add Rental fields	15–16 Oct
○	3.6	Program.cs setup	15–16 Oct
○	3.7	ListCar controller	15–17 Oct
○	3.8	ListCar DTO and its mapper	15–17 Oct
○	3.9	RentalRules service	16–17 Oct
○	3.10	Seed roles, admin account and sample cars	16–17 Oct
○	3.11	Image storage service	16–17 Oct
○	3.12	Branches endpoint	17 Oct
○	3.13	Authentication and authorization with Identity + JWT ◆ M2	17–19 Oct
○	3.14	ChangePassword endpoint	19–20 Oct
○	3.15	Testing with the front end	13–21 Oct
○	3.16	Bug fixing and edits	13–22 Oct
○	3.17	.http file for every endpoint	15–22 Oct

SPRINT 4 — PLANNED

Customer endpoints

23 Oct – 2 Nov · 11 days · 12 cards · names added at sprint start

○	4.1	ListMyRentals controller	23–25 Oct
○	4.2	ListRentals DTO and its mapper	23–25 Oct
○	4.3	CarDetails endpoint (customer)	23–25 Oct
○	4.4	RentalRequest controller	25–27 Oct
○	4.5	RentalRequest DTO and its mapper	25–27 Oct
○	4.6	CancelRental controller	27–29 Oct
○	4.7	UserProfileDto and its mapper	29–31 Oct
○	4.8	GetUserProfile controller	29–31 Oct
○	4.9	RefuseRental controller	29–31 Oct
○	4.10	Testing with the front end	23 Oct – 2 Nov
○	4.11	Bug fixing and edits	23 Oct – 2 Nov
○	4.12	UpdateProfile and UpdateLicense endpoints	31 Oct – 2 Nov

03 · WORK BREAKDOWN, CONTINUED

Sprints 5 to 8: the staff side, then final testing

SPRINT 5 — PLANNED

Admin listing functionalities

3–12 Nov · 10 days · 9 cards · names added at sprint start

5.1	ListCustomers controller	3–5 Nov
5.2	ListCustomer DTO and its mapper	3–5 Nov
5.3	VerifyLicense endpoint (staff)	5–6 Nov
5.4	ListCars controller	5–7 Nov
5.5	AdminListAllRentals controller	7–9 Nov
5.6	ListPayments controller	9–11 Nov
5.7	ListPayments DTO and its mapper	9–11 Nov
5.8	Testing with the front end	3–12 Nov
5.9	Bug fixing and edits	3–12 Nov

SPRINT 6 — PLANNED

Admin core functionalities

13–22 Nov · 10 days · 7 cards · names added at sprint start

6.1	ApproveRental controller	13–15 Nov
6.2	HandOverKeys controller	15–17 Nov
6.3	Rental rejection controller	17–19 Nov
6.4	AddCar controller	19–21 Nov
6.5	AddCar DTO and its mapper	19–21 Nov
6.6	Testing with the front end	13–22 Nov
6.7	Bug fixing and edits	13–22 Nov

SPRINT 7 — PLANNED

Admin endpoints, part 2

23 Nov – 2 Dec · 10 days · 6 cards · names added at sprint start

7.1	EditCar controller	23–25 Nov
7.2	RecordReturn controller	23–25 Nov
7.3	Admin profile endpoints	25–27 Nov
7.4	DeleteCar controller ◆ M3	25–29 Nov
7.5	Testing with the front end	23 Nov – 2 Dec
7.6	Bug fixing and edits	23 Nov – 2 Dec

SPRINT 8 — PLANNED

Final testing

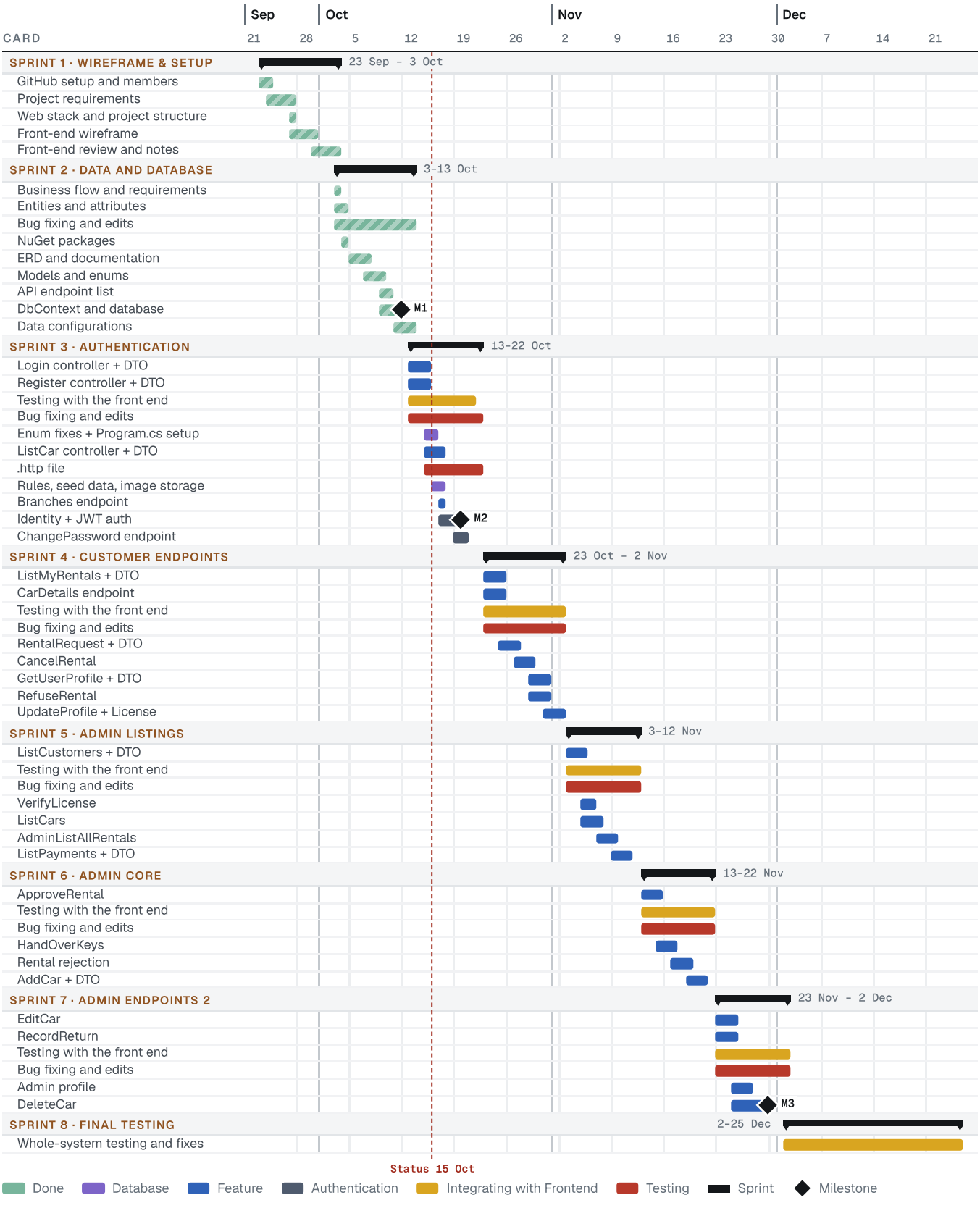
2–25 Dec · 24 days · 1 card · names added at sprint start

8.1	Test the whole system and fix any bugs	2–25 Dec

04 · SCHEDULE

From 23 September to 25 December

Each bar is a Trello card, from its start date to its due date. Cards with the same dates share a row. The attached GanttProject file holds the same chart, one task per card.



04 · SCHEDULE, CONTINUED

Sprints and milestones

Each sprint starts the day the previous one ends, or the day after, so a late sprint moves every date after it. Milestones are the cards with the Milestone label on the board.

SPRINT CALENDAR

SPRINT	FOCUS	DATES	DAYS	CARDS	STATUS ON 15 OCT
S1	Front-end wireframe and project structure	23 Sep - 3 Oct	11	6	DONE
S2	Back-end data and database	3-13 Oct	11	9	DONE
S3	Authentication endpoints and one customer endpoint	13-22 Oct	10	17	0 OF 17 DONE
S4	Customer endpoints	23 Oct - 2 Nov	11	12	PLANNED
S5	Admin listing functionalities	3-12 Nov	10	9	PLANNED
S6	Admin core functionalities	13-22 Nov	10	7	PLANNED
S7	Admin endpoints, part 2	23 Nov - 2 Dec	10	6	PLANNED
S8	Final testing	2-25 Dec	24	1	PLANNED

MILESTONES

REF	DATE	MILESTONE	DONE WHEN	CARD
M1	Sun 11 Oct	DbContext and database ready	The DbContext is set up and the database is created from the models.	Sprint 2: DbContext and database
M2	Mon 19 Oct	Authentication and authorization live	Login and register run on ASP.NET Core Identity with JWT tokens and roles.	Sprint 3: Authentication and authorization with Identity + JWT
M3	Sun 29 Nov	Last app feature built	DeleteCar, the last feature card on the board, is done.	Sprint 7: DeleteCar controller
End	Fri 25 Dec	Final testing complete	The whole system is tested with the front end and the bugs found are fixed.	Sprint 8: last due date on the board

Critical path: data layer (Sprint 2) → authentication (Sprint 3) → customer endpoints (Sprint 4) → staff endpoints (Sprints 5 to 7) → final testing (Sprint 8). The open cards of the current sprint get the first slot in every weekly meeting.

KEEPING THE SCHEDULE IN SYNC

Trello
The source of truth. Dates, names and labels change here first.

→

GanttProject
Wayfare-Schedule.gan: the same cards as tasks, grouped by sprint, with the milestones.

→

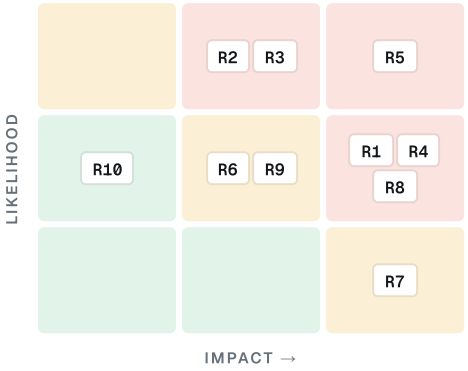
Excel
Wayfare-Planning.xlsx: sprint backlog, sprints, milestones, risk register and team.

> When a card's dates change on Trello, the team leader updates the Gantt chart and the workbook in the same week. Both files are attached to this PDF.

05 · RISK CONTROL

What could go wrong, and what we do about it

We list risks now, while they are cheap to prevent. Each risk is rated for likelihood and impact on a scale of 1 to 3 and gets one concrete action. The team leader checks the register every week; it is also in the attached Excel workbook.



Identify. Anyone can add a risk to the register at any time.

Assess. Rating = likelihood × impact. 6 or more is high, 3 to 4 is medium.

Act. Each risk has one action that starts before the risk happens.

Review. The weekly meeting checks every high risk, and the rest are re-rated at the end of each sprint.

ID	RISK	L	I	RATING	PREVENTION AND FALLBACK
R1	**A team member drops out or is away** 7 people, exams, other courses	2	3	HIGH 6	Whole-team cards carry every name, and notes and code stay in shared places, so another member can pick up any card.
R2	**Scope creep** New features keep arriving	3	2	HIGH 6	The scope is the endpoint list agreed in Sprint 2. New ideas wait for a later sprint and need a change request.
R3	**Front end and back end do not match** Different field names or formats	3	2	HIGH 6	The endpoint list was written in Sprint 2, DTOs keep the front end's field names, and Sprints 3 to 7 each have a Testing with Frontend card.
R4	**Double booking bug** Two customers get the same car	2	3	HIGH 6	Overlaps are checked in the back end when a rental request is saved (Sprint 4), not only on screen, and retested in Sprint 8.
R5	**Schedule slips** Late cards push testing out	3	3	HIGH 9	Card due dates are checked every week. Sprint 8 gives 24 days for final testing and fixes. Low-value features are cut first.
R6	**Poor communication** Messages missed in a big group	2	2	MEDIUM 4	One group chat for quick talk, one Teams channel for decisions. Decisions are written in the notes, not only said.
R7	**Lost code or data** Bad merge, deleted files	1	3	MEDIUM 3	All code lives in the team GitHub repository set up in Sprint 1, with reviewed merges to main. Weekly database backup.
R8	**Security weakness** Passwords and license data	2	3	HIGH 6	ASP.NET Core Identity and JWT in Sprint 3, role checks on every admin endpoint, input validation, retested in Sprint 8.
R9	**Bugs found too late** Testing left to the end	2	2	MEDIUM 4	Every sprint from Sprint 2 has its own bug-fixing card, and testing with the front end starts in Sprint 3.
R10	**Works on one laptop only** Different setups	2	1	LOW 2	The web stack, project structure and NuGet packages were agreed in Sprints 1 and 2. Setup steps go in the README.

06 · TOOLS, COMMUNICATION AND QUALITY

How seven people stay in step

One tool per job, all free. Tool names are links to their sites; the Trello link opens our board.

PLANNING

Trello

The task board and our source of truth. One list per sprint, one card per task, with names, a start date, a due date and labels.

PLANNING

Notion and Excel

Notion for the shared wiki. The Excel workbook holds the sprint backlog, the milestones and the risk register.

`Wayfare-Planning.xlsx, attached`

SCHEDULING

GanttProject

Holds the master Gantt chart, built from the Trello dates. The team leader updates it every week.

`Wayfare-Schedule.gan, attached`

COMMUNICATION

Microsoft Teams

Weekly meeting, channels and file sharing. Official decisions are posted here.

COMMUNICATION

WhatsApp

Quick questions and nudges only. Anything that matters is copied to Teams or the notes.

DOCUMENTATION

Word and PDF

Drafts in Word or Google Docs (the endpoint list lives there), final versions exported to PDF. GitHub keeps the code and its history.

MEETING RHYTHM

MEETING	WHEN	PURPOSE
Weekly planning	`Monday, 30 min, Teams`	Review last week, check due dates on the board, name the next cards, check high risks. The whole team attends.
Mid-week check-in	`Thursday, 15 min`	Everyone shares blockers. We agree hand-offs between areas.
Friday demo	`Friday, 20 min`	Show what works, live or on video. Anything not shown is not counted as done.
Sprint review	`End of each sprint`	Tick finished cards, move unfinished ones to the next sprint, re-rate risks, update the Gantt chart.

Definition of done

- Code reviewed by a second person and merged to **main**.
- Works with the front end, checked in the sprint's testing card.
- Handles empty, loading and error states.
- Due date ticked complete on the Trello card, and noted in the weekly summary.

Change control

1. **Raise.** Anyone fills a short change request.
2. **Assess.** The team leader checks effort, effect on dates and risks.
3. **Decide.** The team votes to approve, delay or reject it.
4. **Record.** The board, the Gantt chart and this plan are updated the same day.

Escalation: if a card is blocked for more than two days, the member tells the team leader. If it is still blocked after a week, the team leader takes it to the weekly meeting and, if needed, to the course instructor.